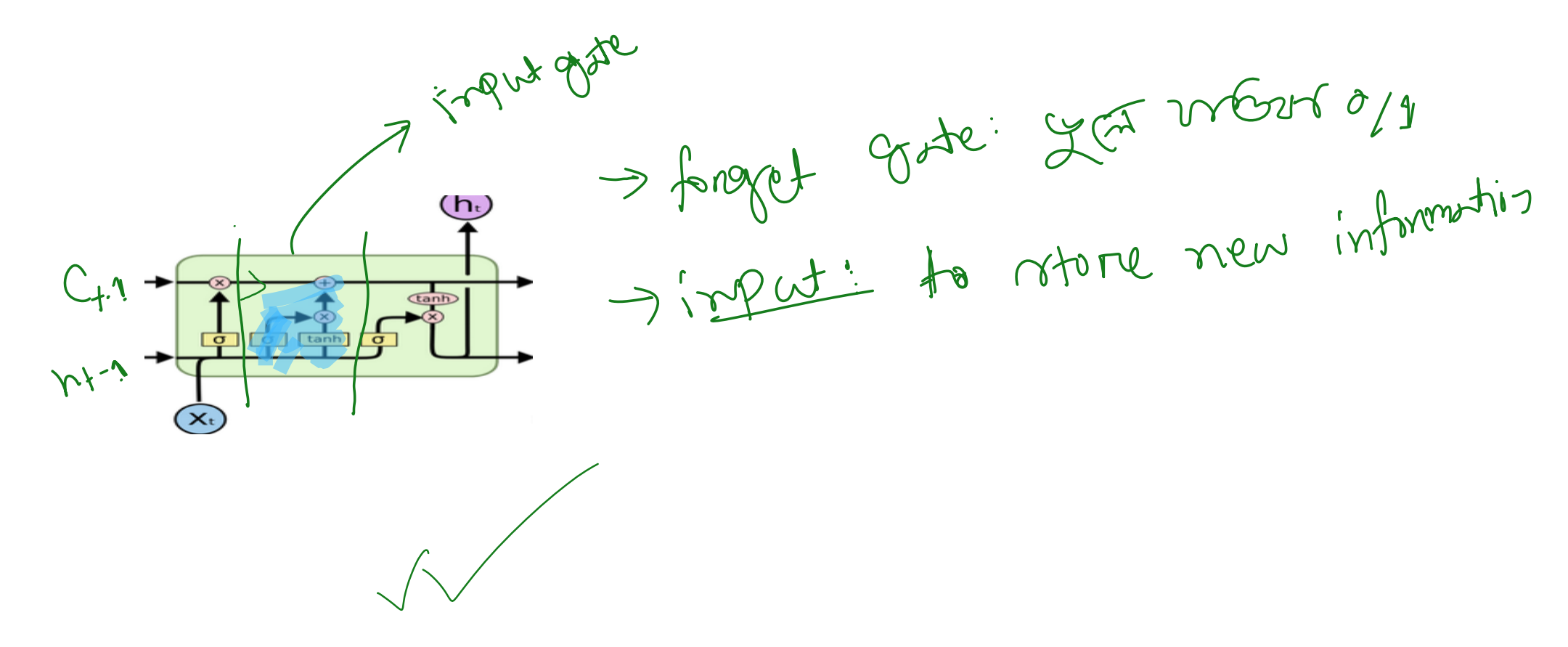




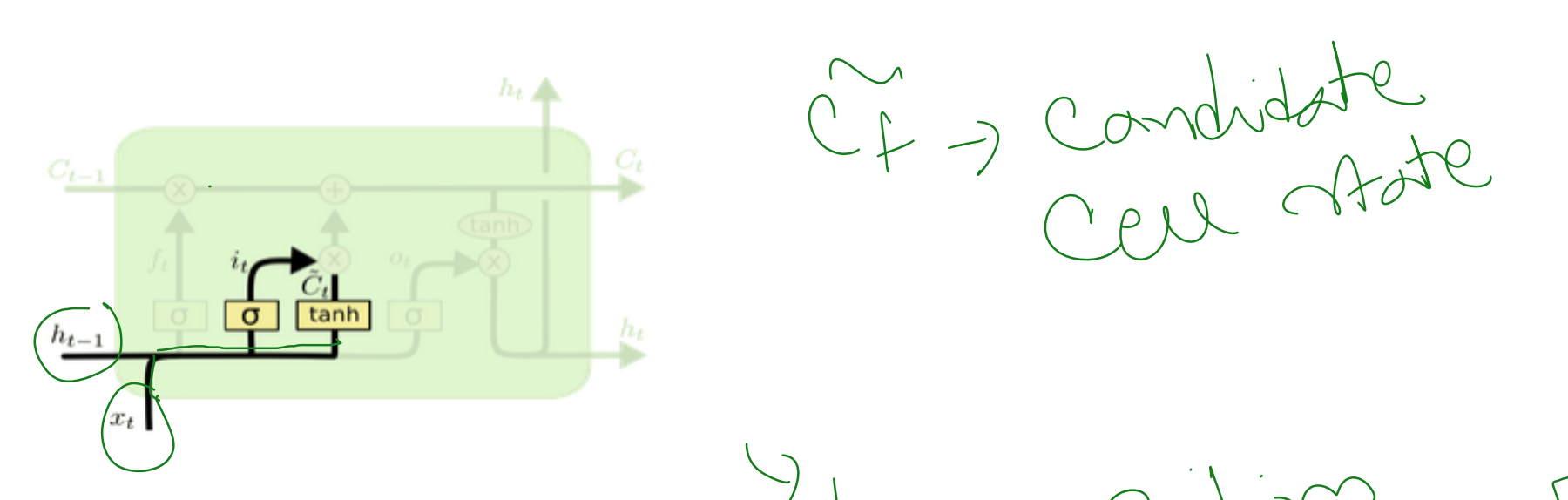
$i_t \rightarrow$
 $C_t \rightarrow$ Candidate Cell State

$\Rightarrow$ calculate i_t and C_t

① $C_t \rightarrow$ candidate cell state

② $i_t \rightarrow$ filtering i_t

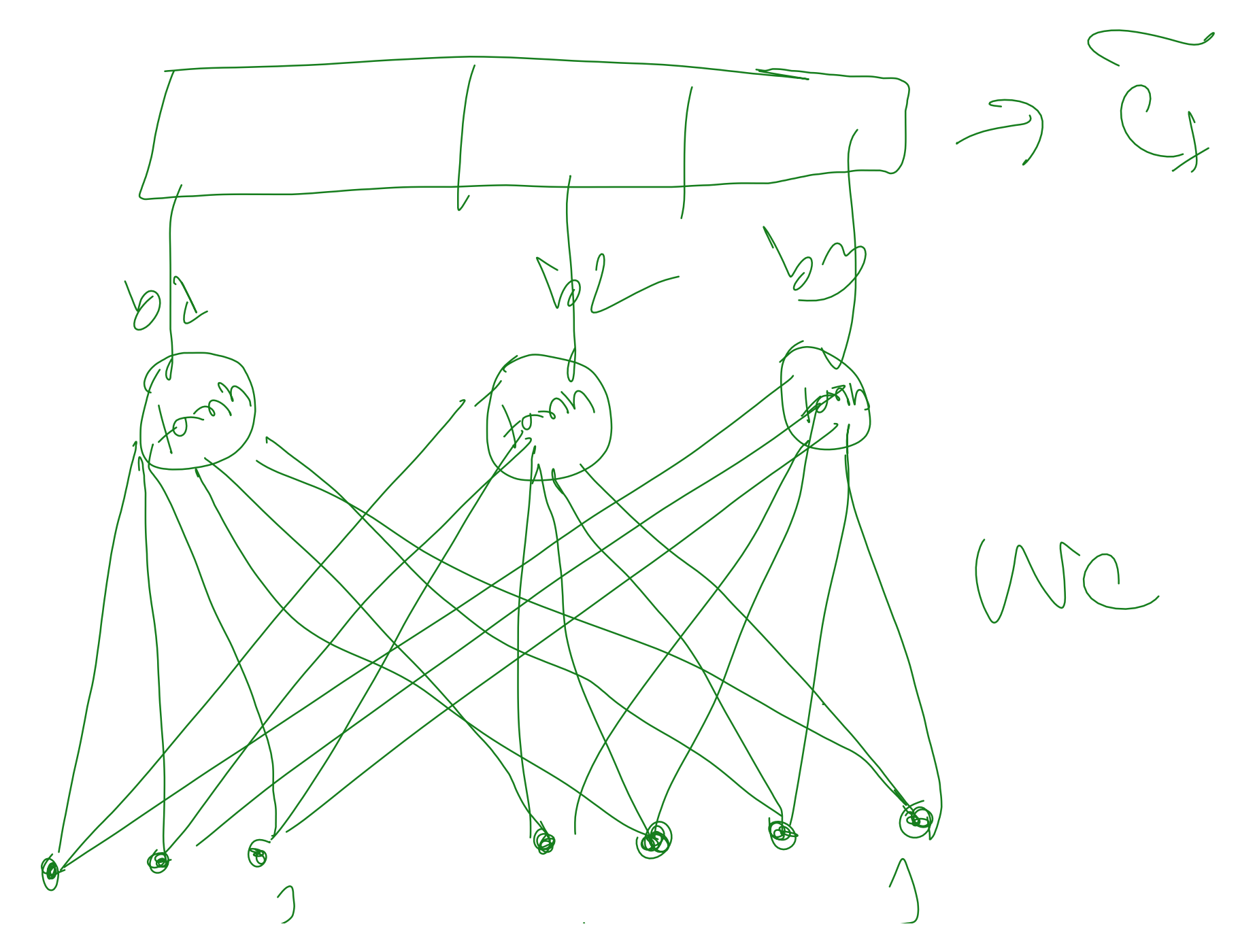
③ $C_t \rightarrow$ cell state



$\tilde{C}_t \rightarrow$ candidate cell state

$\hookrightarrow h_{t-1} = 3 \text{ dim}$
 $\hookrightarrow x_t = 4 \text{ dim}$
 $[h_{t-1}, x_t]$

$\Rightarrow$ neural network layer = 3

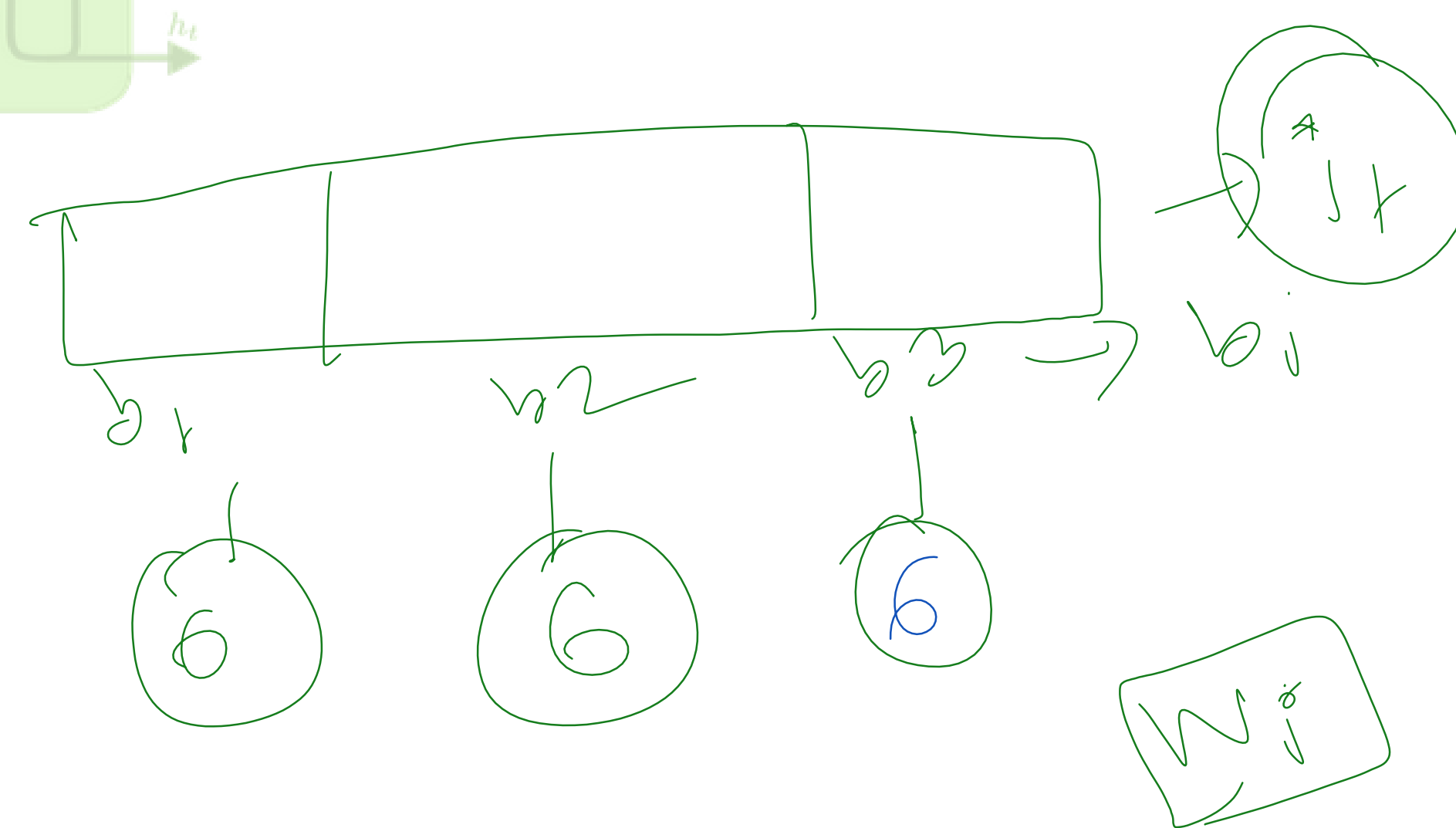
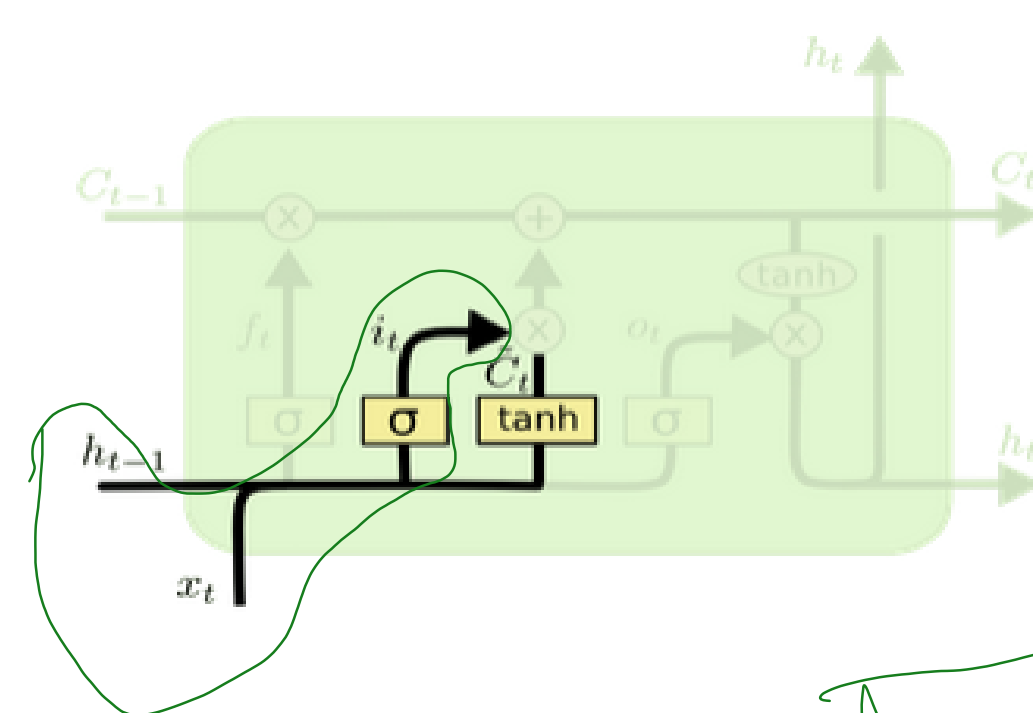




$$\hat{c}_t = \tanh \left(w_c [h_{t-1}, x_t] + b_c \right)$$

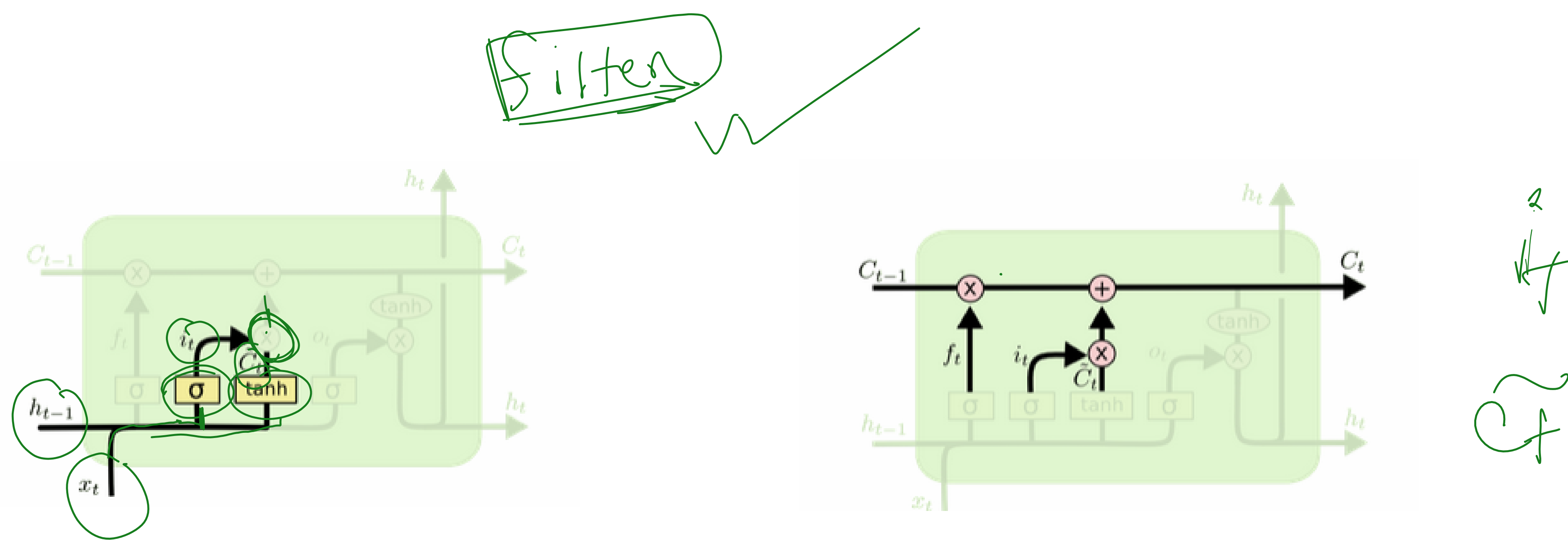
$\downarrow$ $\circlearrowright$ x_{t+1}
 $\circlearrowleft$ h_{t+1}
 $\downarrow$ (w_{t+1})

Calculate I_t



$\circ \circ \circ$
 h_{t-1}
 $\circ \circ \circ \circ$
 $+ x$

$$\hat{i}_t = \sigma \left(w_i [h_{t-1}, x_t] + b_i \right)$$



$$\begin{aligned}
 & i + \tilde{C}_t \rightarrow [4 \ 5 \ 6] \otimes [0 \ 0 \ 0] \\
 & \rightarrow [4 \ 5 \ 6] \quad [1 \ 1 \ 1] \\
 & [0.5 \ 0.5 \ 0.5] \rightarrow [2 \ 2.5 \ 3]
 \end{aligned}$$

$$\begin{aligned}
 & \checkmark \\
 & C_t = f + \tilde{C}_t \rightarrow [4 \ 5 \ 6] \\
 & C_{t-1} = [4 \ 5 \ 6] \rightarrow [1 \ 1 \ 1] \quad [0 \ 0 \ 0]
 \end{aligned}$$

✓ ✓ ✓
✓
[4 5 6]

✓
C₁-1
✓